Tracing Authorial Identity Through Iterative Paraphrasing

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Abstract

Repeated machine paraphrasing can preserve a document's meaning while changing the linguistic patterns that reveal its origin. We study this process in the Ship of Theseus Paraphrased Corpus across three research questions: which text features change first (RQ1), how original-source attribution degrades (RQ2), and whether rewriting systems leave detectable fingerprints (RQ3). Our lexical and semantic baseline uses 3,718 complete chains from Human-origin Yelp and XSum documents; extended structural and attribution experiments use 132,659 complete chains from seven datasets and Human plus six LLM source labels. We find increasing POS and dependency-tree drift across T0–T3, a marked drop in Human-versus-LLM attribution after the first rewrite, and detectable T1 paraphraser signatures (random-forest accuracy 63.82%). Remaining T3 source signals caution against declaring complete identity loss, while detectable paraphraser traces remain relevant to AI-text forensics.

1 Introduction

Large language models can rewrite text while preserving its main meaning, but repeated paraphrasing can change vocabulary, sentence structure, and writing style, potentially removing linguistic patterns that identify the original author or source. The Ship of Theseus paradox asks whether an object remains the same after its components are gradually replaced. In our setting, these components correspond to measurable linguistic cues, including lexical choices, syntactic patterns, and stylistic features. Using the Ship of Theseus Paraphrased Corpus, we compare original texts (T0) with paraphrased versions (T1–T3) generated by ChatGPT, PaLM2 (stored as PaLM in the released corpus), Dipper, and Pegasus. We measure changes in style, word choice, syntax, and semantic meaning, test whether authorship-attribution models can still identify the original source, and examine whether different paraphrasers leave distinct linguistic fingerprints. This problem is relevant to digital forensics, academic integrity, authorship attribution, and AI-generated-text detection. Accordingly, this project addresses three research questions:

RQ1: Which linguistic features are replaced first during iterative paraphrasing? Specifically, how do stylistic features and semantic content decay across T1, T2, and T3?

RQ2: At what paraphrasing iteration does a text lose detectable original-source category information, as measured by the decline in Human-versus-LLM attribution performance?

RQ3: Do different paraphrasing models leave distinguishable linguistic fingerprints that allow the paraphraser to be identified?

2 Background and Related Work

1. Authorship Attribution for Neural Text Generation. In the work done by Uchendu et al. in [1], the authors examined whether machine-generated text can be attributed to its source through same-generator verification, human-vs.-machine classification, and generator identification. They evaluated texts from one human source and eight neural generators, including GPT-2, GROVER, and CTRL, using several classification models. Higher-quality outputs from GPT-2, GROVER, and FAIR more frequently confused the classifiers, showing that attribution becomes harder as generation quality improves.

2. Threads of Subtlety: Detecting Machine-Generated Texts Through Discourse Motifs. In the work done by Kim et al. in [2], the authors investigated whether human and machine writing differ in discourse structure rather than only surface features such as vocabulary and sentence length. They modeled documents using hierarchical parse trees and recursive hypergraphs, extracted discourse motifs, and incorporated them into neural classifiers. Human writing showed greater structural variability and the features improved detection on paraphrased data, but the study did not examine progressive changes across multiple paraphrasing rounds.

3. A Multifaceted Framework to Evaluate Evasion, Content Preservation, and Misattribution in Authorship Obfuscation Techniques. In the work done by Altakrori et al. in [3], the authors argued that authorship obfuscation had often been evaluated using weak identification models and incomplete measures of content preservation and misattribution. They re-evaluated several obfuscation techniques with stronger models under cross-topic conditions and introduced an information-theoretic measure of misattribution harm. A simple back-translation baseline remained competitive across all three evaluation aspects, showing that general-purpose rephrasing can obscure authorial signals while preserving content.

4. Paraphrasing Evades Detectors of AI-Generated Text, but Retrieval Is an Effective Defense. In the work done by Krishna et al. in [4], the authors investigated whether AI-generated-text detectors remain reliable after machine-generated text is paraphrased. They developed DIPPER, a controllable paragraph-level paraphraser, and tested rewritten outputs against DetectGPT, GPTZero, watermarking, and other detectors. DIPPER greatly reduced detection performance while preserving most of the original meaning, showing that paraphrasing can remove detectable AI-writing signals without major semantic change.

5. BERTScore: Evaluating Text Generation with BERT. In the work done by Zhang et al. in [5], the authors studied how to evaluate semantic similarity when texts use different words but express the same meaning. They proposed BERTScore, which compares contextual BERT embeddings rather than exact word or n-gram overlap, and evaluated it on text-generation and paraphrase tasks. BERTScore correlated more closely with human judgments than traditional overlap metrics and was robust to meaning-preserving paraphrases, making it suitable for measuring semantic preservation across T0–T3.

6. A Ship of Theseus: Curious Cases of Paraphrasing in LLM-Generated Texts. In the work done by Tripto et al. in [6], the authors investigated whether a text retains its original authorship after repeated paraphrasing by large language models and other paraphrasing systems. They constructed the Ship of Theseus Paraphrased Corpus, which combines human-written and machine-generated texts from multiple datasets and organizes each document by its source, key, paraphraser, and paraphrasing generation. Their experiments showed that classification performance declines across successive paraphrasing iterations as the rewritten text moves farther from the original author's style. This work directly motivates our source filtering, T0–T3 version mapping, complete-chain construction, and matched-source analysis.

7. On the Evaluation Metrics for Paraphrase Generation. In the work done by Shen et al. in [7], the authors examined whether commonly used automatic metrics can reliably evaluate paraphrase generation. They compared reference-based and reference-free variants of BLEU, ROUGE, METEOR, BERTScore, and BARTScore with human annotations and proposed ParaScore, a family of metrics that explicitly models both semantic similarity and lexical divergence. Their results showed that many commonly used metrics, particularly BLEU and ROUGE, do not consistently align with human judgments. This study motivates our use of BLEU and ROUGE to measure lexical retention together with BERTScore to measure semantic retention, while treating the metrics as complementary rather than directly calibrated indicators.

8. Paraphrase Generation and Evaluation on Colloquial-Style Sentences. In the work done by Sjöblom et al. in [8], the authors studied paraphrase generation for colloquial-style sentences in six languages. They generated paraphrases using RNN and Transformer models trained on the Opusparcus corpus, conducted human evaluations in five languages, and compared the human judgments with BLEU and BERTScore. They found that BLEU correlated reasonably well with human judgments at the corpus level, whereas BERTScore performed better at both the corpus and sentence levels. This finding supports our interpretation of the matched-source results, in which lexical overlap decreases substantially while semantic similarity remains comparatively stable.

9. DetectGPT: Zero-Shot Machine-Generated Text Detection Using Probability Curvature. In the work done by Mitchell et al. in 2023 [9], the authors investigated whether machine-generated text could be detected without training a separate classifier or collecting labeled human and machine-generated examples. To solve this problem, they proposed DetectGPT, a zero-shot method that compares a passage's log probability with the probabilities of semantically similar perturbations, and evaluated it on several domains, including the XSum dataset used in our project. Although DetectGPT maintained an AUROC above 0.8 when approximately 24% of a machine-generated XSum passage was replaced, its performance gradually declined as the amount of revision increased, indicating that repeated rewriting can progressively weaken detectable machine-generated signals.

10. Adversarial Authorship Attribution for Deobfuscation. In the work done by Zhai et al. in 2022 [10], the authors investigated whether authorship-obfuscated texts could still be attributed when the attribution model was aware that an obfuscation method might have transformed the text. To address this problem, they adversarially trained authorship-attribution models using obfuscated examples and evaluated the models under both correct and incorrect assumptions about whether and which obfuscator had been applied. Although adversarial training reduced the effectiveness of existing obfuscation methods from approximately 20–30% to 5–10%, the study examined dedicated authorship obfuscators rather than iterative LLM paraphrasing, leaving open whether authorial signals can be similarly recovered across the T1–T3 paraphrasing chain.

11. Topic or Style? Exploring the Most Useful Features for Authorship Attribution. In the work done by Sari et al. in [11], the authors investigated whether authorship attribution systems rely more on topical information or stylistic features. They analyzed four authorship datasets under different experimental conditions and compared discrete and continuous feature representations. To solve the problem, they evaluated multiple feature types and incorporated the findings into an extended authorship attribution model. Their experiments examined how dataset characteristics affect the usefulness of lexical, stylistic, and topic-related features. The authors showed that feature effectiveness depends strongly on the dataset, and their improved model outperformed previous state-of-the-art systems on two of the four datasets. This work motivates our use of multiple linguistic features rather than relying on a single representation.

12. Idiosyncratic but not Arbitrary: Learning Idiolects in Online Registers Reveals Distinctive yet Consistent Individual Styles. In the work done by Zhu and Jurgens in [12], the authors studied how individual writing styles, or idiolects, can be represented in online text. They focused on whether authors have distinctive but relatively stable stylistic patterns across different online registers. To solve the problem, they trained a neural model using large-scale cross-author comparisons and evaluated it on authorship identification and analogy-based probing tasks. They also used text perturbation experiments to measure the contribution of different linguistic elements to author-specific variation. The results showed that the learned representations captured both qualitative and quantitative differences in individual writing styles. This supports our use of feature-space projections to study how authorial identity moves across iterative paraphrasing stages.

13. TinyStyler: Efficient Few-Shot Text Style Transfer with Authorship Embeddings. In the work done by Horvitz et al. in [13], the authors investigated few-shot authorship style transfer, where a system changes the writing style of a text while preserving its original meaning. To solve the problem, they introduced TinyStyler, which combines an 800-million-parameter language model with pretrained authorship embeddings. The model was evaluated on authorship style transfer and formal-to-informal style transfer using both automatic and human evaluations. The proposed method outperformed strong baselines, including GPT-4, while requiring substantially fewer parameters. However, the study focuses on transferring text toward a target style, whereas our project measures how repeated paraphrasing removes the original style.

14. JAMDEC: Unsupervised Authorship Obfuscation using Constrained Decoding over Small Language Models. In the work done by Fisher et al. in [14], the authors studied authorship obfuscation, which aims to modify text so that the original author

becomes more difficult to identify while the content remains understandable. To solve the problem, they proposed JAMDEC, an inference-time and user-controllable obfuscation method based on constrained decoding with small language models. The method attempts to produce sufficiently large stylistic changes while maintaining content preservation and fluency. Their experiments showed that JAMDEC performed better than previous small-model obfuscation methods and was competitive with much larger language models. This work is relevant to our study because iterative paraphrasing can be viewed as an indirect form of authorship obfuscation.

15. A Large-Scale Computational Study of Content Preservation Measures for Text Style Transfer and Paraphrase Generation. In the work done by Babakov et al. in [15], the authors examined how accurately automatic metrics measure content preservation in text style transfer and paraphrase generation. To solve the problem, they compared 57 evaluation measures across 19 annotated datasets, including lexical metrics, embedding-based metrics, and cross-encoder methods. They also introduced the Mutual Implication Score, which evaluates whether two texts mutually entail each other. The authors found that cross-encoder-based measures generally outperformed traditional alternatives, while the Mutual Implication Score performed particularly well for paraphrase detection. This supports our use of multiple complementary metrics and cautions against interpreting BLEU or ROUGE alone as complete evidence of semantic preservation.

3 Dataset

3.1 Corpus and Paraphrasing Chains

We study the Ship of Theseus Paraphrased Corpus [6]. Each document has an original version T_0, followed by sequential rewrites T_1, T_2, and T_3; each T_g is obtained by paraphrasing T_{g-1}. The seven domains are CMV (discussion/argumentation), ELI5 (explanations), SciGen (scientific text), TLDR (summaries), WP (creative writing), XSum (news summaries), and Yelp (reviews). Source labels include Human and six model-source categories (OpenAI, PaLM, LLAMA, BigScience, Tsinghua, and Eleuther-AI). The four paraphraser families are ChatGPT, PaLM, Dipper, and Pegasus. With Dipper standard/low/high and Pegasus slight/full settings, there are seven paraphraser configurations in total.

The first lexical and semantic baseline focuses on Human-origin Yelp and XSum documents, two contrasting writing domains. Yelp's raw file has 72,847 rows, including 10,781 Human-source rows and 491 unique Human keys; XSum has 70,388 raw rows, including 10,054 Human-source rows and 476 unique Human keys. For the four canonical paraphrasers this subset yields 3,718 complete chains (Table 1), including 3,276 chains from 819 source keys that have all four paraphrasers available (483 Yelp and 336 XSum keys). Using the matched-source subset reduces differences driven by which documents happen to have an available rewrite. In particular, XSum PaLM supplies only 340 complete chains, a data-availability difference rather than evidence of poorer model quality.

Table 1: Complete T0–T3 chain counts for the Human-source Yelp and XSum subset.

Dataset	ChatGPT	PaLM	Pegasus	Dipper	Total
Yelp	488	486	491	491	1,956
XSum	471	340	475	476	1,762
Total	959	826	966	967	3,718

3.2 Alignment, Scope, and Feature Records

The input files contain `source`, `key`, `text`, and `version_name`. We map original and repeated paraphraser version names to T_0–T_3, reshape into one row per dataset–key–source–paraphraser chain, and retain only complete, nonempty four-stage sequences. We check unique chain identifiers and consistency of the original text across paraphrasers. The expanded seven-domain, seven-setting corpus includes Human and all six LLM original-source labels, yielding **132,659 complete chains** and **530,636 stage-level records**. Thus the smaller Yelp/XSum Human-source baseline remains an analysis subset, not an additional independent dataset. The retained chains cover all three rewriting stages for each included original, but complete-chain selection may bias coverage toward documents with available paraphrases.

The extractor parses only 411,463 unique text strings after exact deduplication, then restores the features to all stage-level records. All POS and dependency-depth features are present; 48 version-level rows (0.009%) lack one or more lightweight word-based features owing to tokenization edge cases. In predictive experiments, missing values are imputed using training data alone. The identifier `dataset::key` groups all sources, stages, and paraphraser settings from one document key so they cannot cross training/test partitions; this does not guarantee the absence of repeated documents assigned unrelated keys.

4 Methodology

4.1 Linguistic Feature Extraction

For each version, we calculate 23 numerical features. Five initial stylometric measures are (1) lowercase type–token ratio, (2) average sentence length, (3) population variance of sentence length, (4) Unicode punctuation per 100 word tokens, and (5) average word length. Words use a rule-based Unicode tokenizer retaining internal apostrophes and excluding numbers; sentence boundaries are approximated using final punctuation and line breaks. The expanded feature set adds the 17 universal POS-tag proportions and one dependency-depth measure from the English spaCy parser. We define document depth as the mean, over sentences, of the maximum number of dependency-head links from a token to its sentence root. These measurements capture surface style and syntax, not semantic quality directly.

4.2 RQ1: Style and Content Retention

We compare each T_g ($g = 1, 2, 3$) with the same chain's T_0. BLEU and ROUGE-1/2/L F1 quantify word and sequence overlap; lower values mean fewer matching surface expressions, not necessarily changed meaning. BERTScore F1 [5] is used as a contextual semantic-similarity proxy, complementing rather

than replacing lexical measures [7]. The baseline results used `distilbert-base-uncased` without baseline rescaling; because the saved score files do not record every extraction setting, regeneration metadata should be checked before exact replication.

For each of the five initial style features f, bounded retention is

$$R_f(g) = 1 - \frac{|f(T_g) - f(T_0)|}{|f(T_g)| + |f(T_0)|}, \tag{1}$$

with retention defined as one when both values are zero. The five retentions are averaged without learned weights. The score represents closeness to T_0 under this definition; BLEU, ROUGE, BERTScore, and style retention are *not* directly calibrated measures of a common underlying quantity.

For POS proportions $q_p(T)$ over the 17 tags $p \in \mathcal{P}$, the T0-relative total-variation distance is

$$D_{\mathrm{POS}}(g) = \tfrac{1}{2} \sum_{p \in \mathcal{P}} \left| q_p(T_g) - q_p(T_0) \right|. \tag{2}$$

An increasing value indicates grammatical redistribution. Dependency-depth changes are recorded both as $d(T_g) - d(T_0)$ (direction) and $|d(T_g) - d(T_0)|$ (magnitude). We average each chain-level metric by paraphraser and stage. For Human–LLM structural comparisons, category means are computed within each dataset–paraphraser pairing and then averaged equally over the 49 pairings, limiting domination by the largest dataset.

4.3 RQ2: Origin Attribution and Identity Trajectory

We operationalize an original-source signal as *Human versus LLM provenance*, not an individual's authorship. A class-balanced radial-basis-function SVM ($C = 1$) learns from the 23 features of *only* T_0 texts; the same fitted classifier is evaluated at T_0, T_1, T_2, and T_3. The training-only pipeline uses median imputation and standardization. To guard against document leakage, dataset-stratified 80/20 splitting of distinct `group_id` values (seed 42) creates 2,416 training and 604 held-out groups. There are 15,469 deduplicated original T0 texts in training. The held-out T0 evaluation contains 3,873 unique originals, whereas each paraphrased stage has 26,598 variant-level records. This evaluation therefore mixes a unique-original baseline with pooled later variants and cannot be interpreted as a perfectly paired author-level estimate.

Because the binary source labels are imbalanced, primary metrics are balanced accuracy, macro-F1, and Human-class precision and recall. We derive descriptive 95% intervals using 2,000 bootstrap resamples of held-out document groups, preserving correlation within groups. A large performance drop is evidence of degraded source recognition but is not, by itself, a point where attribution becomes indistinguishable from chance.

We also aggregate features within dataset, document group, source category, and stage before projecting them into a shared two-dimensional PCA space. Median imputation, scaling, and the PCA axes are fitted from training groups. We report Human–LLM centroid separation after averaging source-category centroids equally across the seven domains. A balanced-sample t-SNE plot (2,800 records) offers exploratory support, but PCA centroid distances are the principal quantitative geometric result. Neither plot, without measuring within-group spread, proves compression into a single machine cluster.

4.4 RQ3: Paraphraser Fingerprint Classification

The preliminary fingerprint test identifies the rewriting model from T_1 alone: ChatGPT, PaLM, standard Dipper, or Pegasus(full). The intensity variants Dipper(low/high) and Pegasus(slight) are excluded from this four-label experiment. All seven domains and both Human and LLM originals contribute 74,643 T1 records, of which 59,662 are used for training and 14,981 for the group-held-out test. The same key-level separation yields zero train/test group overlap. Predictors include the 23 structural and style features only; keys, datasets, original-source labels, and paraphraser labels are *not* model inputs. We compare a majority-label baseline, class-balanced standardized logistic regression, and a class-balanced random forest (200 trees, minimum leaf size 3). Imputation is learned on training rows; fixed initial hyperparameters are not tuned on the test set. We report accuracy, macro-F1, confusion matrices, and per-class metrics.

5 Experiments and Results

5.1 RQ1: Which Features Change First?

5.1.1 Lexical and Semantic Retention. Our Human-origin matched-source comparison uses the original Yelp and XSum baseline (Figure 1, Table 2). Across both datasets, BLEU and ROUGE-L decline after T_1 and continue downward through T_3, while BERTScore F1 remains comparatively high. For example, XSum ChatGPT BLEU decreases from 0.262 at T1 to 0.201 at T3, while BERTScore decreases from 0.892 to 0.874; Yelp ChatGPT BLEU decreases from 0.179 to 0.127, while BERTScore changes from 0.869 to 0.845. At T3, Dipper has the lowest lexical overlap in both baseline datasets (BLEU 0.084 on XSum and 0.073 on Yelp). On XSum, PaLM retains BLEU 0.435 and BERTScore 0.919; these measurements characterize the tested outputs rather than a general ranking of paraphraser quality. Across the eight dataset–paraphraser comparisons, BERTScore F1 exceeds ROUGE-L F1, consistent with stronger rewriting of exact word sequences than of the contextual information captured by BERTScore.

Table 2: Mean T3 scores on matched sources (matched Human-source documents).

Data	Model	BLEU	R-L	BERT	Style
XSum	ChatGPT	.201	.447	.874	.882
XSum	PaLM	.435	.652	.919	.908
XSum	Dipper	.084	.245	.823	.882
XSum	Pegasus	.248	.516	.852	.855
Yelp	ChatGPT	.127	.390	.845	.850
Yelp	PaLM	.203	.480	.864	.856
Yelp	Dipper	.073	.250	.814	.856
Yelp	Pegasus	.221	.505	.865	.834

5.1.2 Stylometric and Syntactic Decay. In the same two-domain baseline, sentence-length variance changes most under the bounded

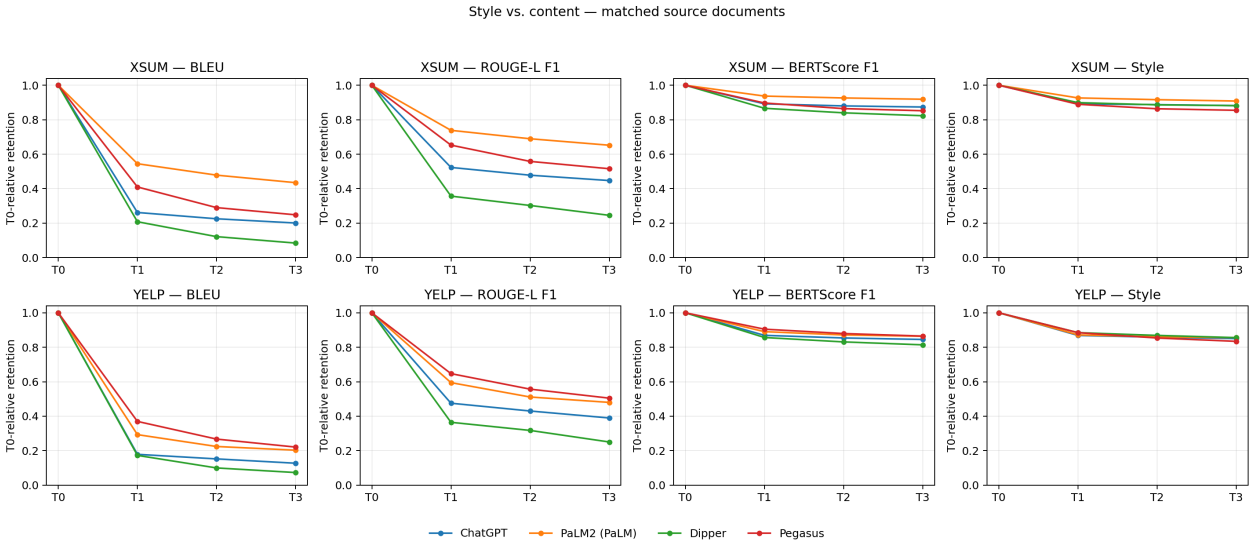


Figure 1: T0–T3 retention trajectories for the matched Human-source Yelp and XSum documents. Lexical, semantic, and style measures are shown on their own descriptive scales.

five-feature style index: T3 retention is 0.681–0.756 on XSum and 0.601–0.663 on Yelp. TTR retention is substantially closer to its T0 value (0.941–0.970 on XSum and 0.960–0.966 on Yelp), as is average word length (0.952–0.985 and 0.929–0.969). These are descriptive properties of a rule-based tokenization and segmentation procedure, not a definitive hierarchy of stylistic durability.

The seven-domain experiment provides deeper evidence on syntax (Figure 2, Table 3). Mean POS drift grows from T1 to T3 for every tested paraphraser setting. Dipper(high) has the largest T3 POS change (0.281), whereas Pegasus(slight) has the smallest (0.072). Dipper(high) also shows the largest T3 absolute dependency-depth shift (1.963); Pegasus(slight) again has the smallest (0.565). Paraphraser strength matters: for Dipper, the high-strength setting produces much more structural change than the low-strength one. Direction is also important: ChatGPT's mean signed depth change is +0.349 at T3, while Pegasus(full) has −1.246. Thus a similar absolute change need not represent a similar syntactic transformation.

Table 3: RQ1: T3 structural changes relative to each chain's T0. Signed depth values clarify the direction of change.

Paraphraser	POS drift	Abs. depth	Signed depth
ChatGPT	.126	1.000	+.349
PaLM	.110	.862	+.102
Dipper	.149	1.025	−.330
Dipper(low)	.096	.867	−.049
Dipper(high)	.281	1.963	−.518
Pegasus(slight)	.072	.565	−.442
Pegasus(full)	.148	1.337	−1.246

The original-source comparison is directionally consistent across matched dataset–paraphraser cells (Table 4). At T3, mean POS drift is 0.132 for Human origins and 0.140 for LLM origins; mean absolute depth differences are 0.973 and 1.106. These differences are small relative to variation among paraphraser settings and may reflect text properties or domain composition rather than a stable Human–LLM rule.

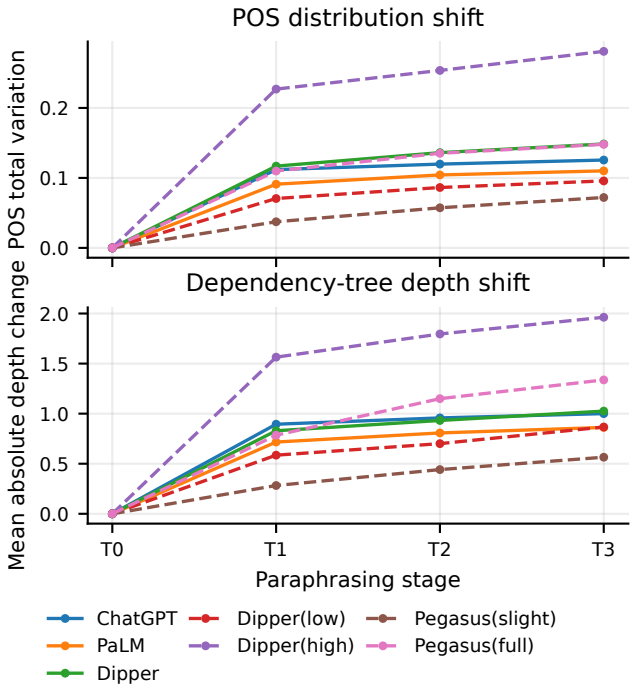


Figure 2: RQ1: mean T0-relative POS total-variation distance (top) and absolute dependency-depth change (bottom) across all seven paraphraser settings. Each mean is computed on aligned complete chains.

Table 4: RQ1: T3 mean structural drift by original source, averaged equally across 49 dataset–paraphraser combinations.

Metric	Human	LLM	LLM−Human
POS total variation	.132	.140	+.008
Abs. dependency depth	.973	1.106	+.133

Individual POS-Tag Frequency Changes. To investigate which grammatical features contribute to POS distribution drift, we compare the mean frequencies of 17 POS tags across T0–T3 for Human-origin texts. Determiners (DET) show the largest increase, from 0.083 at T0 to 0.093 at T3. Proper nouns (PROPN) decrease from 0.049 to 0.040, while adverbs (ADV) decrease from 0.044 to 0.036. Most changes occur between T0 and T1, followed by smaller shifts in later iterations. These results suggest that paraphrasing changes individual grammatical patterns, although the average frequency differences are relatively small.

5.2 RQ2: When Does Original-Source Attribution Decline?

5.2.1 The Attribution Decay Curve. Balanced accuracy of the T0-trained SVM declines from 0.800 at T0 to 0.669 at T1, then to 0.646 at T2 and 0.631 at T3 (Figure 3, Table 5). Macro-F1 declines from 0.693 to 0.553, 0.539, and 0.535. The largest adjacent loss is therefore between T0 and T1 (macro-F1 −0.140); the additional T1–T3 decline is only 0.018. This provides evidence that the first rewrite is the most disruptive stage for the tested original-source distinction.

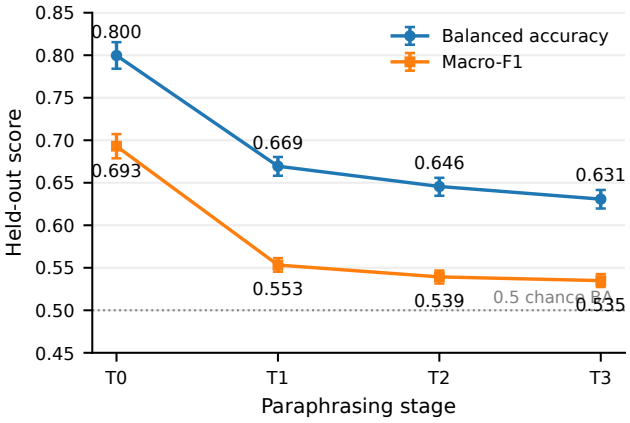


Figure 3: RQ2: T0-trained Human/LLM attribution evaluated on held-out T0–T3 records. Error bars are 95% group-cluster bootstrap intervals. The dotted 0.5 reference applies only to balanced accuracy.

The T3 balanced-accuracy 95% interval (0.620–0.642) remains above the binary 0.5 chance reference. Thus **no definitive point of no return is established through T3** for Human-versus-LLM attribution. Accordingly, the observed decline represents a weakening of source-category provenance rather than demonstrated loss of individual-author identity. The classifier still retains some original-category signal, even though its predictions become less reliable. Human recall falls from 0.838 at T0 to 0.655 at T3, and precision

Table 5: RQ2: held-out Human-versus-LLM attribution. The test set has 604 document groups at each stage.

Measure	T0	T1	T2	T3
Accuracy	.774	.624	.613	.614
Balanced accuracy	.800	.669	.646	.631
Macro-F1	.693	.553	.539	.535
Human recall	.838	.735	.693	.655
Human precision	.394	.252	.239	.232

from 0.394 to 0.232; the overrepresentation of LLM-origin rows makes these class-specific results essential. At T3, dataset-level balanced accuracy ranges from 0.573 on TLDR to 0.716 on CMV, indicating that pooling also conceals important domain variation. Comparisons across stages require caution because T0 originals are deduplicated while transformed stages contain repeated variants.

5.2.2 Identity Trajectory in Feature Space. A complementary PCA analysis shows movement of Human and LLM category centroids as texts are paraphrased. The first two components account for 24.45% and 9.91% of standardized feature variation (34.36% combined). The dataset-balanced distance between the Human and LLM centroids decreases from 0.275 at T0 to 0.146 at T1, 0.142 at T2, and 0.139 at T3, about a 49.4% reduction overall (Figure 4). As in attribution, most of the measured change occurs after the first rewrite. Smaller centroid

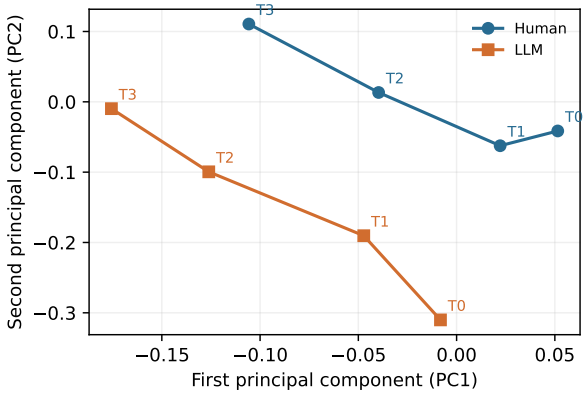


Figure 4: RQ2: dataset-balanced Human and LLM centroid trajectories in the shared PCA coordinate system. The plot depicts movement of group means, not the full distribution of documents.

distance is a geometric description of reduced separation of category means, not a guarantee that documents converge into a dense "machine region." PCA omits much of the 23-dimensional variation and the 2D plot does not quantify within-class dispersion; t-SNE also cannot establish that stronger conclusion by visual inspection alone. The two RQ2 analyses are therefore complementary rather than interchangeable: classification tests a learned boundary, while PCA summarizes a lower-dimensional projection.

5.3 RQ3: Can We Identify the Paraphraser?

5.3.1 Four-Class T1 Classification. The random forest obtains 63.82% accuracy and 0.623 macro-F1 on 14,981 held-out T1 versions,

compared with 59.12% and 0.578 for logistic regression and 25.85% and 0.103 for the majority baseline (Table 6). The improvement over the baseline supports the presence of distinguishable stylistic information about the rewriting system, although it does not imply that every paraphrase is identifiable.

Table 6: RQ3: T1 paraphraser classifier results on 14,981 held-out records (59,662 training records).

Classifier	Accuracy	Macro-F1
Majority baseline	.259	.103
Logistic regression	.591	.578
Random forest	.638	.623

5.3.2 Errors and Candidate Linguistic Signatures. The confusion matrix (Figure 5) shows that Pegasus is the easiest class to recover (recall 0.770), followed by ChatGPT (0.729), Dipper (0.668), and PaLM (0.353). Among 3,416 PaLM test texts, 943 are predicted as ChatGPT and 781 as Dipper. RQ3 therefore requires per-class reporting rather than treating 63.82% as uniform success. Logistic regression coefficients associate ChatGPT with average word length, Dipper with mean sentence length, and Pegasus with punctuation frequency; PaLM shows positive associations with punctuation and sentence-length variation. These are conditional correlations in a multicollinear feature set, not reliable causal explanations. The classification task evaluates four canonical paraphraser classes at T1, excluding additional intensity variants and all T2–T3 stages.

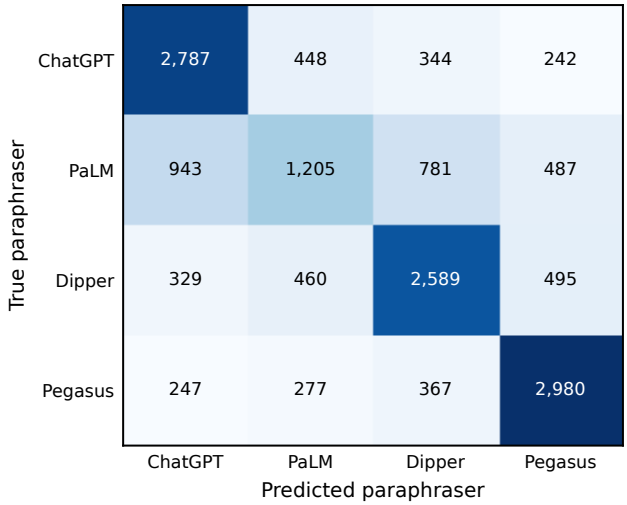


Figure 5: RQ3: random-forest held-out confusion matrix for the four T1 paraphrasers. Rows are ground-truth labels and columns are model predictions.

6 Discussion

6.1 Origin Cues and Transformation Cues

The three RQs reveal related but distinct information in a rewritten document. RQ1 establishes feature-level changes in surface overlap, style, and grammatical structure. RQ2 shows that the Human–LLM origin category becomes harder to recover, particularly after one paraphrase, while remaining above a chance benchmark at T3. RQ3 shows that even when origin cues weaken, the transformation mechanism can leave its own recognizable style. This is not a contradiction: the labels refer to different processes, and the experiments have different training targets and sampling procedures. Related work on attribution, obfuscation, and style transfer [1, 3, 11–14] motivates examining both signals instead of reducing forensic identity to one score.

6.2 What the Results Do and Do Not Establish

The strongest supported pattern is an early decline in origin-category classification performance, alongside accumulated structural drift across later stages. We do not infer that individual authorship is irrecoverable at T1, because the current classifier predicts only binary original-source category; neither do we infer that the PCA points collapse into one machine cluster. Lexical overlap and BERTScore are on different distributions, and the bounded retention index is defined by a chosen formula, so higher BERTScore numbers should not be read as a precisely calibrated "slower semantic decay rate." These distinctions are important when interpreting the Ship of Theseus analogy: the text can acquire a new surface form and still retain some original-category cues and identifiable transformation traces.

7 Limitations

The lexical/semantic baseline covers Human-origin Yelp and XSum only; its BERTScore settings and domain composition are not directly equivalent to the expanded seven-domain structural experiment. Complete-chain filtering excludes unavailable rewrites and may bias samples. POS distributions and dependency depth depend on the English parser, rule-based tokenization influences the five initial style features, and the five-feature retention measure treats different measurements equally without validation against human judgments. BERTScore is a useful semantic proxy but not an entailment or factual-preservation guarantee [15].

The RQ2 model separates Human versus LLM origins, not distinct writers or individual generating models. Its highly imbalanced source categories, the T0-versus-T1–T3 evaluation-weighting difference, and any hidden related texts across document keys can affect the estimated decay. The 95% group-bootstrap intervals summarize sampling uncertainty within the chosen split, not all sources of modeling uncertainty. PCA retains only 34.36% of variance in two dimensions and its centroid measurement does not quantify compression or clustering. RQ3 uses only a single key-disjoint T1 test split and four canonical classes; its feature coefficients may reflect correlated measures, and its findings are not yet validated across external domains or later paraphrase rounds.

8 Conclusion and Future Work

Our results provide initial answers to all three research questions: repeated paraphrasing changes lexical and structural features at different rates (RQ1); Human–LLM attribution declines most after T1 but retains detectable signal at T3 (RQ2); and four canonical

paraphrasers are partially identifiable from their T1 linguistic signatures (RQ3). RQ2 therefore concerns source-category provenance, not individual authorship. Before the final report, we will validate semantic retention on matched documents, test individual POS and domain effects with paired uncertainty, strengthen RQ2 with author- or generator-specific labels where supported, compare matched-original stage weighting, quantify PCA within-category spread, and extend RQ3 to later iterations and cross-domain generalization. These analyses will help distinguish transformed surface form from truly lost provenance information.

Code and Reproducibility. The modular code, notebooks, result tables, and plots are available at https://github.com/xiaoyancai/Ship-of-Theseus-Paraphrased-Corpus-CS6120-Team4, branch `project-update-2`. Large processed files are distributed separately with setup instructions in the README; complete reproduction requires those files or regeneration from the original data and a compatible parsing environment.

References

[1] Adaku Uchendu, Thai Le, Kai Shu, and Dongwon Lee. 2020. Authorship Attribution for Neural Text Generation. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Association for Computational Linguistics, 8384–8395. https://doi.org/10.18653/v1/2020.emnlp-main.673.

[2] Zae Myung Kim, Kwang Lee, Preston Zhu, Vipul Raheja, and Dongyeop Kang. 2024. Threads of Subtlety: Detecting Machine-Generated Texts Through Discourse Motifs. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Association for Computational Linguistics, 5449–5474. https://doi.org/10.18653/v1/2024.acl-long.298.

[3] Malik Altakrori, Thomas Scialom, Benjamin C. M. Fung, and Jackie Chi Kit Cheung. 2022. A Multifaceted Framework to Evaluate Evasion, Content Preservation, and Misattribution in Authorship Obfuscation Techniques. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Association for Computational Linguistics, 2391–2406. https://doi.org/10.18653/v1/2022.emnlp-main.153.

[4] Kalpesh Krishna, Yixiao Song, Marzena Karpinska, John Wieting, and Mohit Iyyer. 2023. Paraphrasing Evades Detectors of AI-Generated Text, but Retrieval Is an Effective Defense. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*. Curran Associates, Inc., 27469–27500. https://doi.org/10.52202/075280-1195.

[5] Tianyi Zhang, Varsha Kishore, Felix Wu, Kilian Q. Weinberger, and Yoav Artzi. 2020. BERTScore: Evaluating Text Generation with BERT. In *Proceedings of the 8th International Conference on Learning Representations (ICLR 2020)*. OpenReview.net. https://openreview.net/forum?id=SkeHuCVFDr.

[6] Nafis Irtiza Tripto, Saranya Venkatraman, Dominik Macko, Robert Moro, Ivan Srba, Adaku Uchendu, Thai Le, and Dongwon Lee. 2024. A Ship of Theseus: Curious Cases of Paraphrasing in LLM-Generated Texts. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Association for Computational Linguistics, 6608–6625. https://doi.org/10.18653/v1/2024.acl-long.357.

[7] Lingfeng Shen, Lemao Liu, Haiyun Jiang, and Shuming Shi. 2022. On the Evaluation Metrics for Paraphrase Generation. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Association for Computational Linguistics, 3178–3190. https://doi.org/10.18653/v1/2022.emnlp-main.208.

[8] Eetu Sjöblom, Mathias Creutz, and Yves Scherrer. 2020. Paraphrase Generation and Evaluation on Colloquial-Style Sentences. In *Proceedings of the Twelfth Language Resources and Evaluation Conference (LREC 2020)*. European Language Resources Association, 1814–1822. https://aclanthology.org/2020.lrec-1.224/.

[9] Eric Mitchell, Yoonho Lee, Alexander Khazatsky, Christopher D. Manning, and Chelsea Finn. 2023. DetectGPT: Zero-Shot Machine-Generated Text Detection Using Probability Curvature. In *Proceedings of the 40th International Conference on Machine Learning (ICML 2023)*, 24950–24962. Proceedings of Machine Learning Research. https://proceedings.mlr.press/v202/mitchell23a.html.

[10] Wanyue Zhai, Jonathan Rusert, Zubair Shafiq, and Padmini Srinivasan. 2022. Adversarial Authorship Attribution for Deobfuscation. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL 2022), Volume 1: Long Papers*, 7372–7384. Association for Computational Linguistics. https://aclanthology.org/2022.acl-long.509/.

[11] Yunita Sari, Mark Stevenson, and Andreas Vlachos. 2018. Topic or Style? Exploring the Most Useful Features for Authorship Attribution. In *COLING 2018*, 343–353. https://aclanthology.org/C18-1029/.

[12] Jian Zhu and David Jurgens. 2021. Idiosyncratic but not Arbitrary: Learning Idiolects in Online Registers Reveals Distinctive yet Consistent Individual Styles. In *EMNLP 2021*, 279–297. https://aclanthology.org/2021.emnlp-main.25/.

[13] Zachary Horvitz, Ajay Patel, Kanishk Singh, Chris Callison-Burch, Kathleen McKeown, and Zhou Yu. 2024. TinyStyler: Efficient Few-Shot Text Style Transfer with Authorship Embeddings. In *Findings of EMNLP 2024*, 13376–13390. https://aclanthology.org/2024.findings-emnlp.781/.

[14] Jillian Fisher, Ximing Lu, Jaehun Jung, Liwei Jiang, Zaid Harchaoui, and Yejin Choi. 2024. JAMDEC: Unsupervised Authorship Obfuscation using Constrained Decoding over Small Language Models. In *NAACL 2024*, 1552–1581. https://aclanthology.org/2024.naacl-long.87/.

[15] Nikolay Babakov, David Dale, Varvara Logacheva, and Alexander Panchenko. 2022. A Large-Scale Computational Study of Content Preservation Measures for Text Style Transfer and Paraphrase Generation. In *ACL Student Research Workshop 2022*, 300–321. https://aclanthology.org/2022.acl-srw.23/.